Girls Build Bots — Cadette Cybersecurity Safeguards

Badge-aligned workshop guide for robotics-team-led outreach

Level: Cadettes (Grades 6–8) · **Badge:** Cybersecurity — Cybersecurity Safeguards **Series:** Cybersecurity (Basics · Safeguards · Investigator) **Format:** One 2-hour session · **Ratio:** 1 mentor per 5–6 Scouts **Who runs it:** A FIRST robotics team (FLL / FTC / FRC) leading the session. A physical robot is a nice bonus but **not required** — every activity runs on printed materials and free browser tools.

> **Troop Leaders:** This curriculum is designed to *align with* the Cadette Cybersecurity badge series. Official badge requirements are in your Girl Scout Volunteer Toolkit (VTK). Verify completed activities against the official steps before awarding badges. Badges are purchased through your council shop.

The through-line: understanding, protecting, investigating

At the Cadette level, Safeguards gets real: attackers mostly don't hack computers — they hack *people*. Social engineering, urgency, impersonation. The defense is a calm, skeptical mind plus a few strong technical habits.

Vocabulary: social engineering, phishing, impersonation, authentication, two-factor (2FA), password manager, privacy audit.

Cybersecurity Safeguards (2 hours)

Badge theme: Real protection strategies — authentication, social-engineering defense, and deliberately securing your digital life.

Materials

- **Made-up** credential examples for the authentication lab (never real passwords — state this rule aloud)
- Mentor-made "social engineering" scenario deck: printed DMs/emails/texts with escalating subtlety (fake giveaway → fake friend-in-trouble → fake "school office" request → one perfectly legit message mixed in)
- "Privacy audit" checklist worksheet (account types, what each knows about you, sharing settings to review with a parent)
- Optional: NOVA Cybersecurity Lab for reinforcement

Activity 1 — Two Locks Beat One (Authentication Lab) (25 min)

Quick passphrase refresher (length beats cleverness; three-random-words method) using invented examples only — then level up to **authentication**: how does a system know you're

you? Something you *know* (passphrase), something you *have* (a code on a phone — **2FA**), something you *are* (fingerprint). Demo the logic: a stolen passphrase alone fails against 2FA. Mention **password managers** as the grown-up tool: one strong master passphrase, unique passwords everywhere else. Each Scout leaves with a written personal authentication plan (made-up examples, real methods).

Activity 2 — Social Engineering Defense (40 min)

The heart of the session. Work the scenario deck in teams: for each message, identify the **manipulation lever** — urgency ("in the next 10 minutes!"), authority ("this is the school office"), emotion ("your friend is stranded"), reward ("you've been selected"). Then the counter-move: **slow down, verify through a separate channel** (call the real number, ask the real friend in person), never act on pressure. The deck's planted legit message matters: the skill is discrimination, not blanket paranoia. Score: which team correctly sorts the full deck *and* names each lever?

Activity 3 — Privacy Audit Workshop (30 min)

Each Cadette works the audit checklist against her own digital life *from memory* (no logging in at the workshop): what accounts exist, what does each know, who can see what, what would she change? Deliverable: a top-three action list to complete at home **with a parent** — tighten one setting, enable 2FA somewhere, retire an unused account. Frame it as taking inventory of your own house keys.

Activity 4 — Defense Drill (15 min) + Wrap (10 min)

Rapid-fire: mentors read scenarios; teams call the lever and the counter-move. Closing: each Scout names the manipulation lever she thinks would most likely have worked on her *before* today — honest self-awareness is the strongest safeguard in the room.

> **Robot moment (optional):** Share for 60 seconds how the team protects accounts and code with unique passphrases and 2FA — and why "someone urgently needs the team login" is a message you verify, never obey.

Completion Checklist — Cybersecurity Safeguards

By the end of the session, to earn the badge, each Scout can: - [ ] Explain authentication factors and why 2FA defeats a stolen password - [ ] Identify the manipulation lever in a social-engineering attempt - [ ] Apply the verify-through-another-channel counter-move - [ ] Complete a privacy audit and commit to three concrete improvements

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Mentor Guide (for the robotics team)

- **Never touch real credentials.** All examples invented; no logging into anything during the session. Say the rule aloud — it *is* the lesson.
- **You don't need a robot for this.** The "robot moment" is a 60-second add-on.

- **Make scenarios age-real.** Game-currency scams, fake giveaway DMs, “friend” asking for a code — their world, not banking fraud.
- **Include the legit message.** Blanket suspicion isn’t the skill; discrimination is. Debrief why the real one was believable.
- **Confidence, not fear.** The closing tone: “you now see the levers — they don’t work as well on someone who sees them.”

📣 Share Your Impact — required for teams and troops using this curriculum

If you use Girls Build Bots materials to run a workshop, please complete all three:

1. **Credit us.** Tell your Scouts and audience that the curriculum comes from Girls Build Bots, and credit “Girls Build Bots” anywhere you share about the event.
2. **Tag us.** Post about your workshop and tag **@girlsbuildbots** on Instagram.
3. **Report your results.** Email **girlsbuildbots@gmail.com** with: the number of girls who participated, the number of badges earned, and a sentence or two of feedback on how it went.

This is how the program grows and how we measure our impact — your report becomes part of the Girls Build Bots story and helps the next troop and team.

Created for the “Girls Build Bots” Girl Scout Gold Award project. Original curriculum aligned to the Cadette Cybersecurity badge series; official requirements © Girl Scouts of the USA, available to troop leaders in the Volunteer Toolkit.